App Dev Project Report

1. Student Details.

Name: Santhosh S K
Roll Number: BS 22F3000732
Email: 22f3000732@ds.study.iitm.ac.in

About Me:
I am a student in the IIT Madras BS Degree Program with interest in web development and database management systems.

2. Project Details.

Project Title: Placement Portal Application

Problem Statement:
To build a **Placement Portal Application** web application that allows **Admin (Institute), Company, and students** to interact with the system based on their roles.

Approach:

The application was developed using Flask as the backend web framework and SQLite as the database. The database is managed using SQLAlchemy ORM, which allows interaction with the database using Python classes.

The system is designed with three main roles:

Admin (Institute):

The admin manages the system and approves company registrations. The admin dashboard displays statistics such as the total number of students, companies, placement drives, and applications.

Company

Companies can register on the portal and wait for admin approval. After approval, companies can create placement drives and view applications submitted by students.

Student

Students can register on the platform, view available placement drives, apply for drives, and track their application status.

The application follows a Model–View–Controller (MVC) style architecture:

- Models: Database tables defined using SQLAlchemy
- Views: HTML templates rendered using Jinja2
- Controllers: Flask route functions handling application logic

The database tables are created programmatically using SQLAlchemy models when the application starts.

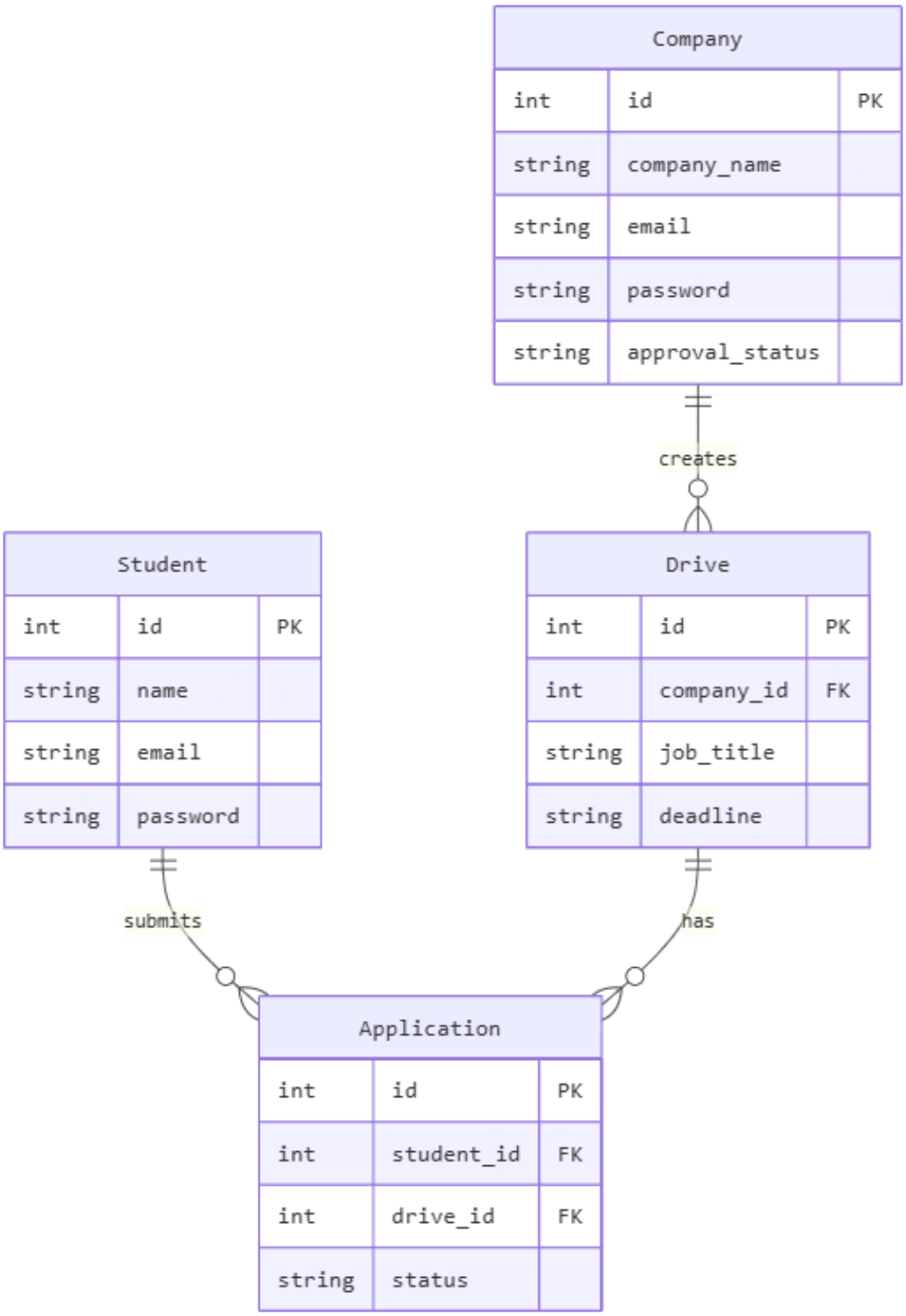
3. AI/LLM Declaration

No AI or Large Language Models (LLMs) were used during the development of this application. The entire application was implemented manually using Flask, SQLAlchemy, HTML, and CSS.

4. Frameworks and Libraries Used

Technology	Purpose
Flask	Backend web framework
Flask-SQLAlchemy	ORM integration with Flask
SQLAlchemy	Database ORM for managing tables and queries
SQLite	Lightweight relational database
Jinja2	Template engine for dynamic HTML rendering
HTML	Structure of web pages
CSS	Styling of the user interface
Bootstrap	Responsive UI design

5. Database Schema / ER Diagram



•

6. API Resource Endpoints

Endpoint	Method	Description
/	GET	Redirects to student registration page
/register_student	GET, POST	Student registration
/register_company	GET, POST	Company registration
/admin_dashboard	GET	Displays admin dashboard
/approve_company/	GET	Approve company registration
/reject_company/	GET	Reject company registration
/create_drive	GET, POST	Create placement drive
/view_drives	GET	View all placement drives
/apply_drive/<drive_id>	GET	Apply for a drive
/student_dashboard	GET	View student applications
/company_applications/<drive_id>	GET	View student applications for a drive

Implemented Features

Student Module

- **Student registration**
- **View available placement drives**
- **Apply for placement drives**
- **Prevent duplicate applications**
- **View application status**

Company Module

- **Company registration**
- **Admin approval workflow**
- **Create placement drives**
- **View student applications**

Admin Module

- **Dashboard with statistics**
- **Approve or reject company registrations**
- **Monitor all students, companies, drives, and applications**

7. Video presentation

Drive link

https://drive.google.com/drive/folders/1KIPW_gnSQDO7aS8I5_yMNnPjNHFi0UUh?usp=drive_link